

## TASK 01 – Writing Better Prompts

### *Prompt Engineering Internship*

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In this task, three vague or ineffective prompts are rewritten into improved, more effective versions by adding clarity, specificity, and an appropriate tone. Each improved prompt is compared with the original to understand how the level of detail in a prompt changes the quality and usefulness of an AI response.

### Topic 1: Healthy Eating

**Vague Prompt:** Tell me about healthy eating.

#### Output (Vague Prompt)

Healthy eating means consuming a balanced diet with fruits, vegetables, proteins, and whole grains while avoiding too much sugar and processed food. It helps maintain energy levels and overall well-being.

#### Improved Prompt:

Act as a nutrition coach. Give me 5 practical healthy-eating tips for a busy college student with a limited budget and little time to cook. For each tip, include one simple reason why it works. Use a friendly, encouraging tone and keep each tip under two sentences.

#### Output (Improved Prompt)

- **Plan Simple Meals:** Cook basics like rice, dal, and vegetables in bulk on weekends — it saves time and keeps you from relying on instant food.
- **Keep Fruit Visible:** Store fruit on your desk instead of snacks — you're more likely to reach for what you can see.
- **Choose Whole Grains:** Swap white bread or maida items for whole wheat or millet options a few times a week — they keep you full longer for less money.
- **Drink Water First:** Have a glass of water before meals — it's free, and it helps you notice real hunger versus boredom.
- **Batch-Cook Protein:** Boil a batch of eggs or soak chana for the week — a quick protein source beats skipping meals when you're rushed.

## Analysis

The vague prompt returns a generic definition of healthy eating with no connection to the reader's situation. The improved prompt defines a persona (nutrition coach), audience (busy, budget-conscious student), output count, structure, and tone. The resulting tips are specific, actionable, and realistic for the intended reader.

## Topic 2: JavaScript Function

**Vague Prompt:** Write a JavaScript function.

### Output (Vague Prompt)

Here is a simple JavaScript function that adds two numbers: `function add(a, b) { return a + b; }`

### Improved Prompt:

Write a beginner-friendly JavaScript function called `checkPassword` that takes a string and returns whether it is 'Weak', 'Medium', or 'Strong' based on length and the presence of numbers and special characters. Add short comments explaining the logic and show one example call with its output.

### Output (Improved Prompt)

```
function checkPassword(password) { // Track how many conditions the password
satisfies let score = 0; if (password.length >= 8) score++; // length check if
(/[0-9]/.test(password)) score++; // contains a number if
(!/@#$%^&*]/.test(password)) score++; // contains a special character if (score ===
3) return "Strong"; if (score === 2) return "Medium"; return "Weak"; } // Example
call console.log(checkPassword("hello@123")); // Output: Strong
```

## Analysis

The vague prompt leaves the AI to guess what kind of function is wanted, producing a trivial example. The improved prompt specifies the function name, purpose, logic, comments, and an example call, so the AI returns working, well-explained code that matches an actual use case.

### Topic 3: Resume Writing

**Vague Prompt:** Give me resume tips.

#### Output (Vague Prompt)

Keep your resume short, use bullet points, highlight your skills and experience, and proofread it before submitting.

#### Improved Prompt:

Give me 5 specific resume-writing tips for a final-year engineering student applying for their first software internship, with no prior work experience. Focus on how to present academic projects and coursework as strengths. Keep each tip within two sentences and use a professional but encouraging tone.

#### Output (Improved Prompt)

- **Lead With Projects:** Since you lack work experience, place your academic and personal projects near the top, right after your summary — recruiters look here first for practical skills.
- **Use Action Verbs:** Start each bullet with a verb like 'Built,' 'Designed,' or 'Optimized' instead of 'Responsible for' — it shows ownership of the work.
- **Quantify Where Possible:** Mention numbers such as 'reduced load time by 30%' or 'handled a dataset of 10,000 rows' — even estimates make your impact concrete.
- **Match the Job Description:** Mirror 2–3 keywords from the internship posting in your skills section — this helps you pass automated resume screening.
- **Keep It to One Page:** Trim unrelated coursework and keep only the most relevant skills — a focused one-page resume is easier for recruiters to scan quickly.

#### Analysis

The vague prompt produces generic, widely available advice that ignores the reader's actual situation. The improved prompt specifies the audience (a student with no experience), the challenge to address (presenting projects as strengths), a tip count, sentence limit, and tone — leading to advice that is specific and directly usable.

## Key Learnings from Task 01

- **Specificity Matters** – A specific prompt gives the AI clear direction and reduces the chance of a generic or irrelevant answer.
- **Define the Audience** – Naming who the response is for helps the AI adjust vocabulary, tone, and depth appropriately.
- **Set the Format** – Asking for bullet points, code, or tables makes the output easier to scan and use.
- **Add Constraints and Scope** – Word limits, counts, and required details keep the response focused instead of open-ended.
- **Persona and Tone** – Assigning a role (coach, developer, career advisor) and a tone shapes how the AI frames its answer.

## Conclusion

This task shows that the quality of an AI's response depends heavily on how the prompt is written. Vague prompts tend to produce broad, generic answers, while prompts with a clear audience, format, and constraints consistently produce more useful and relevant results. Writing better prompts is therefore less about asking more questions and more about asking one well-defined question.